

Concrete & Cement

Concrete

Air Content, Air Void System Parameters

The AVA (Air Void Analyzer)

AVA is used to measure the air-void parameters (spacing factor and specific surface) of samples of fresh air-entrained concrete. Samples are taken after concrete is placed.

With the AVA, the air-void structure is measured after the concrete is placed but while it is still fresh, thereby providing timely information of the spacing factor and the specific surface of the air void system in the cement paste of the in-place concrete. Timely information is important, because practice has shown that the air-void structure created by air entraining agents can change easily during construction; for example, due to the type and dosage of normal or high-range water-reducing admixtures, by changes in sources of cementitious materials, by pressure influences during pumping of concrete, by high hydrostatic pressure, or by over vibration. The testing time is 25 minutes or less.



Concrete Air Meters

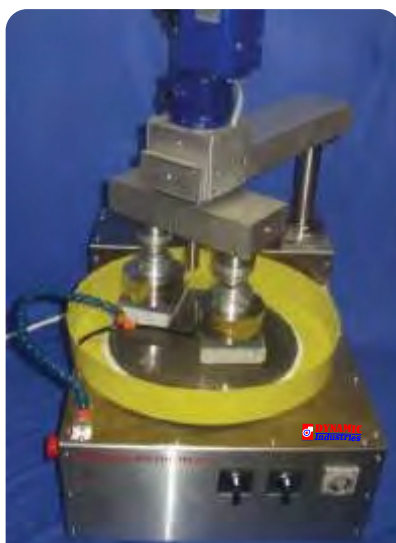
Low maintenance air meter, which uses no moving parts inside the chamber. Pressure is released into the base by an external, brass, quick-release T-valve. The base is machined inside and out for easy cleaning. Chamber and cover are one solid component. No more bottom gasket leaks. Furnished with all necessary accessories for calibration and operation, plus plastic toolbox-style carrying case.

Several types of air meters are available and information can be provided on request.

Rapid Air

RapidAir is an image analysis system for automatic determination of the air content in hardened concrete. The measured parameters of the air-void structure are total air content, spacing factor, and specific surface.

The RapidAir measurement is done within 12 min for linear traverse analysis and within 30 min for modified point count. This should be compared with a time of 4 to 6 hours normally required for manual analysis using a light microscope. Graphical presentation of the air-void distribution and the raw data are also available.



Petro Planner

The PetroPlanner is a state-of-the-art lapping and polishing machine for preparing lapped/polished plane surfaces for Air Void Analysis according to ASTM C457 or EN 480-11. Petrographic examination of polished fluorescent epoxy impregnated surfaces and polishing of thin sections for SEM / EDS examination.

The PetroPlanner is a simple and durable design with a low demand for maintenance and a long service life. Its compact design and low space requirements are useful for laboratory facilities.

The PetroPlanner grinding system, with its different grinding slurries, maximizes the success of such grinding and polishing jobs. The procedure can be supplemented by impregnation with an acetone based hardener that is applied before each grinding step. It is having simple & easy operation, with proven track record of 20 years of excellent results.

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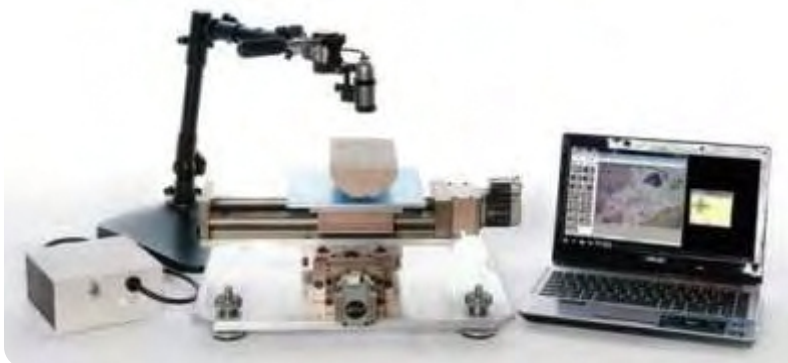
Petrothin

PetroThin is the ultimate thin section preparation machine. Thin sections are used extensively by petrographers to study the microstructure of concrete. The examination of thin sections of concrete and related materials is one of the most powerful methods for determining and characterizing the water-cement (w/c) ratio, mixture proportions, air-void structure and freezing of fresh concrete etc.



Linear Traverse Machine

The Linear Traverse Machine is used to facilitate the counting of microscopic voids in concrete. Computer software controls the motorized specimen table with solid-state positioning motors. The software controls the traverse patterns and distances between measurements based on test sample parameters. Test results can be exported to Microsoft Excel for analysis and printing. Unit includes: Digital microscopic video camera with integral light source, computer-controlled motorized specimen table with solid-state positioning motors, software CD with controller and multi-adjustable camera stand. Computer is not included.



Penetration Testing

Concrete Pocket Penetrometer

Lightweight, spring-reaction type concrete penetrometer for field and lab evaluation of the initial set of concrete mortar, based on ASTM C403. Penetration plunger has a 1/20 sq. in. tip area. Plunger is steadily pushed into the mortar to a 1 in. depth, as indicated on the shaft, at periodic time intervals. Penetrometer's calibrated range is 0-700 psi. Resistance in psi is indicated on the scale. The term "initial set" is the semi-hardened, partially hydrated condition of the concrete beyond which it can no longer be worked.



Acme Penetrometer

Hydraulic reaction-type apparatus for determining the setting time of concrete with slump greater than zero by testing mortar sieved from the concrete mixture. It also determines the effects of variables such as temperature, cement, mixture proportions, additions and admixtures upon the time of setting and hardening of concrete. The penetrometer's design makes it easy to operate, being more efficient, with a longer gear rack. All needles are one length so settings may remain the same.



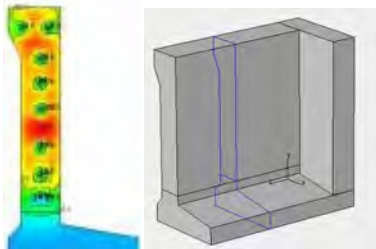
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Heat of Hydration, Cracking Potential

HeatBox

The purpose of the HeatBox is to determine the heat of hydration of a concrete mixture, which may be used as input information for a thermal analysis program. It is one of the components of a complete thermal control system.

The HeatBox determines the heat of hydration of a concrete mixture based on the principle of semi adiabatic calorimetry. A specimen of fresh concrete is placed at the center of an insulated container. A portion of the total heat generated by hydration raises the temperature of the concrete specimen and a portion is absorbed by the container and transferred to surrounding air. The amount of heat absorbed by and transferred through the container can be established by appropriate temperature measurements of the insulation material or it can be determined by numerical simulation provided the thermal conductivity and heat capacity of the insulation are known.



Be4Cast

Be4Cast is an advanced software package for simulating temperature evolution and stress development in concrete structures during early-ages. The software allows modeling different construction methods for a given structure in order to arrive at an optimal solution for reducing the risk of early-age cracking due to thermal gradients and thermal shrinkage. Be4Cast is based on heat transfer in 3-dimensions, which permits more accurate simulation of heat transfer in a structural element of any shape. insulation are known.

Autogenous Shrinkage

Auto-Shrink

The Auto-Shrink system measures the unrestrained autogenous shrinkage of a specimen of cement paste or mortar cured under sealed conditions (ASTM C1698, Test Method for Autogenous Strain of Cement Paste and Mortar). Auto-Shrink permits evaluation of the relative autogenous shrinkage potential of different cementitious systems. Excessive autogenous shrinkage may lead to microcracking that increases the permeability of concrete.

The digital displacement gauge is fixed firmly to the frame with a blunted hex screw. The remote Control is mounted on the side of the gauge. A lock function for the remote control is provided. With Auto-Shrink, it is possible to measure the time dependent deformation of many different specimens simultaneously over periods of weeks or even years.



Rheology and Self Consolidation Concrete

ICAR Rheometer

The ICAR Rheometer provides, for the first time, a low-cost and simple to operate instrument that can be used for: Research and development to characterize the influence of new materials on concrete rheology. The ICAR Rheometer is designed to characterize the static yield stress, the dynamic yield stress and plastic viscosity of the concrete.

The ICAR Rheometer software performs all the necessary functions like operates the driver, records the torque, computes test results, and stores data. For simplicity, the entire program is operated from a single screen. The user defines the test geometry and provides the test parameters to run the flow curve test. A simple press of the "Start" button initiates the tests, which takes less than 1 minute to complete.



J-Ring Test Set, Smooth Rod

The J-ring test can be used to determine the passing ability of self-consolidating concrete. It is applicable for laboratory use in testing different concrete mixtures for passing ability or can be used in the field as a quality control test. Designed for durability, the set includes a slump cone, J-Ring with smooth rods, and steel base plate with engraved rings to measure flow distance.



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L-Box Flowability Test

Method used to determine flow rates and passability of SCC in confined spaces, as well as an evaluation of filling, passing ability and segregation resistance. Test box is comprised of concrete reservoir, slide gate, an obstacle grid comprised of 3 bars and a test basin. Includes metal strike-off bar and instructions.

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V-Funnel Flowability Test

Designed to evaluate the segregation resistance of self-consolidating concrete through the flowing speed from a funnel. Applicable to concrete with aggregates of 1" (25mm) maximum size. Stainless steel construction with 10L capacity. Upper edge is smooth and reinforced and the outflow orifice is equipped with a sealable valve. Includes polyethylene box to collect discharge and a 36" (914mm) long straight edge to level concrete prior to testing.

Ball Penetration Apparatus (Kelly Ball)

Used to test the consistency of concrete using the penetration of a half sphere into plastic concrete. Apparatus consists of 30 lb. (14kg) cylinder with hemispherically shaped bottom and handle. Stirrup or frame guides handle and acts as reference for measuring depth of penetration. Concrete may be tested as placed in the forms prior to any manipulation or in a suitable container.

Static Segregation Column Mold

Used to determine the potential for static segregation of self-consolidating concrete by measuring the coarse aggregate content in the top and bottom portions of a cylindrical specimen. This test can be used as a quality control determination for levels of coarse aggregate segregation, especially for deep section applications like walls or columns. The column includes a stable base, three column sections and two sliding separator blades, which allow the column to be disassembled while capturing the concrete sample in the column sections.

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Strength - Early age, Maturity, Consistency

LOK-TEST : - The LOK-TEST system is used to obtain a reliable estimate of the in-place strength of concrete in newly cast structures. LOK-TEST provides an accurate estimate of in-place strength because the peak pullout force has a well-defined correlation to compressive strength measured using standard cylinders or cubes.

It is useful for determining whether in-place concrete strength is sufficient for early application of loads, such as due to formwork removal, application of prestressing. Determining whether the in-place strength is sufficient for terminating curing and thermal protection. Evaluating the quality of the critical cover layer protecting the reinforcement in the finished structure.



Multi-Channel Maturity Meter:- Humboldt Maturity Meters provide a predictable strength determination of cast-in-place concrete based on ASTM standard C1074-98. These units utilize inexpensive, disposable, T-type thermocouple wire with quick-connect jacks, which can be embedded directly into a concrete structure to measure temperature at timed intervals. The H-2680 and H-2682 have four channels, all four channels may be used simultaneously and provide the maturity number calculation, instant readout and temperature history on a menu driven alphanumeric display.

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COMA-Meter (Concrete Maturity-Meter)

The COMA-Meter (Concrete Maturity-Meter) is used to measure the maturity of newly cast concrete at a depth of 80 mm from the surface for estimating the compressive strength at an early age using a pre-established strength-maturity relationship. Timing of pullout testing with LOK-TEST for early-age strength measurement & evaluating the effective in-place curing temperature.



HeatWatch

HeatWatch is a hardware and software system to measure the temperatures, determine temperature differences, and calculate maturity values at critical locations during hardening of concrete in a structure.

HeatWatch also allows strength estimation at early ages, provided the strength-maturity relationship for the concrete mixture used in the structure has been established and entered into the HeatWatch computer program. HeatWatch can be used to verify predictions of temperature development in a newly cast member based on using a thermal analysis program.



TMS

TMS is a system to measure temperatures (T), calculate maturity (M) values at critical locations and estimate in-place strength (S) at early ages, based on a provided strength-maturity relationship. TMS provides remote monitoring capabilities. Alarms stored in the TMS software alert the operator if preset temperatures or temperature differences have been exceeded. The TMS also permits automatic control of cooling or heating measures.



Consistency Vibrating Table Test

Used for determining the consistency of stiff to extremely dry concrete mixtures like those used in roller-compacted concrete mixtures. Density of the specimens is determined by determining the mass of the consolidated specimen and dividing by its volume. The unit is comprised of a vibrating table, which can be bolted to a floor or substantial base slab. The test mold is 9.5" x 7.75" ID with handles for easy movement and is locked into place on the base with positioning tabs and wing nuts.

Concrete Strength - Lab

Compression Testing Machines

Hydraulic Compression Testing Machines are available in the capacities of 1500KN, 2000KN, 3000KN, 4000KN and 5000 kN. The choice of machine capacity should be made according to size and strength of specimens to be tested in the laboratory. However, since concrete is being produced with higher performance, it is always a good idea to choose a machine which has a higher capacity to that indicated by present day calculations. Dual ram machines are also available which allow testing with two different capacities: 2000/300 kN and 3000/300 kN. These allow higher accuracies in both higher and lower range of testing. The machines are available as simple digital display units OR fully computerized, strain controlled systems. Computerised models with feedback system and automatic load pace regulation enable the entire test to be controlled via the software.



High stiffness Frames

All of our machines are characterized by their high flexural rigidity on all three axes. This is particularly true of the four column models but is also a feature of our welded, enbloc frames thanks to the use of H or tubular section beams which, optimize the stiffness/weight ratio.

Test Platens

The test platens are made in compliance with the flatness, hardness and parallelism criteria of the latest international standards. Minimum surface hardness: 550 HV 30 or, upon request 600 HV (EN 772/1). A wide range of testing platens for blocks and distance pieces are available as accessories.

Hydraulic power

Automatic and computerized compression machines are actuated by means of control consoles that use our "Silent & Cold Power" technology. The semi-automatic machines still use our traditional,

on-board power unit. The latter contains two pumps: a high capacity/low pressure pump used during fast approach procedure for securing the specimen is securely fitted between the test platens and a low capacity/high pressure pump that is used during the actual testing procedure.

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Universal Flexural Strength Testing Device

For concrete beams with centre-point loading method and third-point loading method. The device comprises a lower element with two bearers (one fixed and one floating/rotating bearer). Adjustable span at 300 mm and at 450 mm, an upper element with two floating/rotating bearers fixed in a floating system. Adjustable span at 100 mm and at 150 mm. One of the upper bearers can be removed to perform the centre-point test.

Indirect tensile stress testing device for cylinders Ø 100, 150, 160 mm

For the indirect tensile testing ("Brazilian" test) of concrete cylinders with diameters 100, 150 and 160 mm. The test requires the sample to be taken to failing point by the application of compressive force to the two generating lines of the cylinder.

Indirect tensile stress testing device on soil-cement samples

For the indirect tensile testing ("Brazilian" test) of soil cement specimens 6" x 7"



Compressometer-Extensometer

Compressometer-extensometers are used to determine Poisson's ratio and Young's modulus during a compression test. This device contains a third, center yoke with a hinge dividing it into two equal segments. The middle yoke is hinged to permit rotation of the two segments of the yoke in the horizontal plane. Indicator gives deformation readings. Second indicator is furnished for compressometer section. Unit measures changes in length and diameter.



Compressometer

The compressometer is used for evaluating the chord modulus of elasticity (Young's modulus) of concrete cylinders while undergoing compression testing. The Compressometer includes two cast aluminum-alloy yokes, mounting and central points, stainless steel control rods. Models are available with a dial gauge, as well as with a digital indicator or a LSCT transducer. Digital indicators and LSCT models can be used with data acquisition systems through the use of our miniLoggers.

Concrete Beam Tester

Self-contained shock-proof portable concrete beam tester accurately and easily determines flexural strengths of 6 x 6" cross-section test beams. Hydraulically driven unit uses center-point loading method with continuous readings to the break and retains maximum reading to eliminate lost data. Gauge resets to zero for repeat tests. Lightweight aluminium unit has dual registration of modulus of rupture between 15,000 lbf. and 0-6,700 kgf.



Cylinder End Grinder

This automatic cylinder end grinder quickly grinds specimen ends plane and parallel prior to compression tests. Test cylinders is easy to install on the table and requires no assembly. The machine is mounted on wheels for easy moving. Safe access to components allows for easy maintenance. The grinder's aluminum frame and stainless steel exterior ensure both resistance to corrosion and light weight.

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Concrete Strength - In-place/ NDT

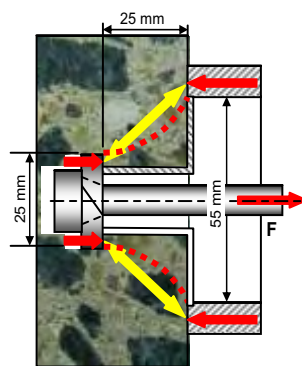
LOK-TEST

The LOK-TEST system is used to obtain a reliable estimate of the in-place strength of concrete in newly cast structures. LOK-TEST provides an accurate estimate of in-place strength because the peak pullout force has a well-defined correlation to compressive strength measured using standard cylinders or cubes.

It is useful for determining whether in-place concrete strength is sufficient for early application of loads, such as due to formwork removal, application of prestressing. Determining whether the in-place strength is sufficient for terminating curing and thermal protection. Evaluating the quality of the critical cover layer protecting the reinforcement in the finished structure.

CAPO (Cut and Pull Out) Test

CAPO test measures the force required to pullout an embedded metal insert from a concrete specimen/ structure. The structure is subjected to a slowly applied load and measures the actual compressive strength properties of concrete. CAPO Test can help determine the in-situ strength of structure, strength of fire-damaged structures among others ensuring safety of the structure. Conforms to ASTM C900, BS 1881:2007, EN 12504-03 and is accredited by the Indian Road Congress (IRC)



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Concrete Test Hammers

Designed for non-destructive testing of concrete. The hammer is pressed against the concrete structure and the rebound values are displayed on a sliding scale. These values can then be somewhat correlated to compressive strength of concrete. Recording type and Silver/ Digital version are also available.



Crack Monitoring

CORECASE

CORECASE is a portable light weight coring rig for quickly obtaining drilled cores that are accurate in diameter, have straight sides, and are perpendicular to the surface. In addition to determining in-place compressive strength, drilled cores can also be used for the Verification of flaws identified by NDT methods. Preparing a partially drilled core for conducting a pull-off test with BOND-TEST rapid chloride penetration bulk conductivity for petrographic analysis. The end result is a quickly drilled core that has a smooth surface, accurate diameter, and drilled perpendicular to the surface.



CMD (Crack Monitoring Device)

The purpose of CMD (Crack Monitoring Device) system is to monitor the opening and closing of a surface crack as well as relative lateral movement between the two sides of the crack as a function of time. With the CMD, a three-point rosette, in the shape of an equilateral triangle with 50-mm side lengths, is bonded to the surface. Two of the rosette points are positioned parallel to the crack, and the third one is positioned on the opposite side of the crack approximately half the distance to the base line.

The measurements are made manually using a caliper, or they can be made electronically using displacement sensors (LVDTs). The CMD also has the option for remote monitoring from the office by means of a GSM cellular phone modem.

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Essential, Digital Crack Monitoring Kit

This set contains all the essential components needed to set up a crack monitoring system using precision digital calipers. This kit includes a crack width gauge to measure the width of cracks prior to monitoring, a digital caliper, a pack of stainless steel discs and crack record sheets. Also included is an illustrated manual with full instructions, which describes best practice methods for using the calipers.



Professional, Digital Crack Monitoring Kit

This set contains all that is needed to set up a crack monitoring system using precision calipers. This kit includes a crack width gauge to measure the width of cracks prior to monitoring, a digital caliper with data port, a pack of stainless steel discs, a pack of caliper marks, a pair of corner discs and crack record sheets

Concrete Crack Data Logger, Wifi

The wireless crack data logger provides a method to monitor cracks and linear displacements in concrete structures, featuring an integrated data logger, which measures both displacement and ambient temperature. This crack detector measures cracks by using a rotary, precision potentiometer, which is driven by the winding/unwinding of a stainless steel wire with an 80mm stroke. The wireless crack data logger downloads data to a PC via a wireless connection using a graphic user interface on a PC computer.

Integrity Testing and Condition Assessment



PUNDIT Lab Ultrasonic Pulse Velocity Test

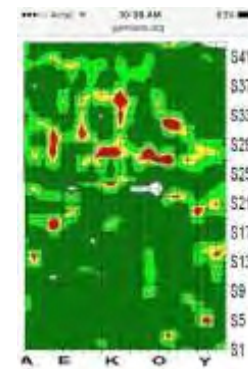
The PUNDIT (Portable Ultrasonic Nondestructive Digital Indicating Tester) Lab is used to measure the propagation speed of a pulse of ultrasonic longitudinal stress waves. The ultrasonic pulse velocity (UPV) that is determined can be used for evaluating the uniformity of concrete within a structural member and the depth of surface-opening cracks.

The PUNDIT Lab and the PUNDIT Lab+, both instruments are capable of investigating the structural integrity of concrete, ceramics and refractory, timber, and other materials

s'MASH –Impulse Response Test

For a long time, users of NDT systems have wished for a rapid, easy to use method for rapid screening of the integrity of structures. The s'MASH impulse-response test system fulfills this wish. The idea is to quickly screen a structure for flaws and identify suspect areas for subsequent detailed investigation.

With the s'MASH, rapid evaluation can be conducted for: Detecting voids beneath concrete slabs in highways, spillways, and floors. Detecting curling of slabs on ground. Locating delaminations and honeycombing in bridge decks, slabs, walls and large structures such as dams, chimney stacks, and silos. Detecting debonding of asphalt or concrete overlays and repair patches from concrete substrates.



DOCTer – Impact Echo Test

The DOCTer is a versatile, portable system based on the impact-echo method. The impact-echo method is based on monitoring the periodic arrival of reflected stress waves and is able to obtain information on the depth of the internal reflecting interface or the thickness of a solid member.

It can be used for measuring the thickness of pavements, asphalt overlays, slabs-on-ground and walls and detects the presence and depth of voids and honeycombing.

It can be used for evaluating the quality of grout injection in post-tensioning cable ducts and detects debonding of overlays and patches. It is useful to measure the depth of surface-opening cracks and estimation of early-age strength development (with proper correlation).

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EyeCon

EyeCon is a portable hand-held instrument for flaw detection and thickness measurement. It is based on the ultrasonic pitch-catch method and uses an antenna composed of an array of dry point contact (DPC) transducers, which emit shear waves into the concrete. It is a dry point contact shear-wave transducers with ceramic wearing tip with rechargeable batteries.

EyeCon can be used for measuring thickness and to detect of voids in grouted tendon ducts. Its useful in detection of voids, honeycombing in concrete members as well as the poor quality bond in overlays and repairs. EyeCon allows the user to display the signal as recorded originally (left side) or as a rectified signal.



MIRA Tomographer

The MIRA Tomographer is a state-of-the-art instrument for creating a three-dimensional (3-D) representation (tomogram) of internal defects that may be present in a concrete element. MIRA is based on the ultrasonic pitch-catch method and uses an antenna composed of an array of dry point contact (DPC) transducers, which emit shear waves into the concrete.

MIRA has been used successfully for the thickness measurement, detection of voids in grouted tendon ducts. Detection of voids and honeycombing in concrete members and delaminations. Detection of voids behind tunnel linings and below slabs on ground..



DK-5000

The DK-5000 determines the resonant frequency of prismatic or cylindrical specimens in accordance with the impact resonance method described in ASTM C215. The impact resonance method is a simple test that determines the resonant frequency very quickly. Resonant frequency testing can be used for Monitoring damage as a result of exposure to accelerated weathering, such as cycles of freezing and thawing in accordance with ASTM C666/C666M,

Covermeters / Rebar Locators

CoverMaster

Covermeters are used to locate reinforcing bars and metal cable ducts in concrete structures, for measure the cover depth of reinforcement, for estimating the size of reinforcing bar and to locate other metal objects embedded in concrete. CoverMaster instruments are based on the pulse-induction technique. A repetitive current pulse is applied to the coils in the search head. It includes standard search head, cable, carrying case, and instruction manual and rechargeable battery pack.



Profometer

The Rebar Locator features a universal probe that allows you to measure both shallow and deep ranges at the push of a button. This gauge will locate, size and show bar orientation, as well as indicate concrete cover. In areas where adjacent and parallel bars affect readings, the operation can include a correction to isolate bars for more accurate diameter determinations. The operator can also include a correction factor for determining cover depth in congested bar arrangements. Highly accurate measurements and single-probe design makes this rebar locator an easy-to-use workhorse. Includes processing and report generation software.



Profoscope

The Profoscope is a versatile, fully-integrated rebar detector and cover meter with a unique real-time rebar visualization, allowing the user to actually "SEE" the location of the rebar beneath the concrete surface to a maximum depth of 180 mm and can determine rebar diameter to a depth of 64mm. The Profoscope is a light device that allows the user to operate this rebar detector with one hand making the task of locating rebars a simple and efficient process.

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Corrosion Evaluation & Service Life

GalvaPulse – Corrosion Analyser:- The GalvaPulse is used to measure the corrosion rate of reinforcement by measuring polarization resistance using the galvanostatic pulse technique, for monitoring corrosion activity in reinforced concrete structures.

The GalvaPulse is a easy to operate, lightweight, handheld equipment. It's a portable system including calibration unit and a check block with embedded stainless steel and corroding black steel bars. The test results are in 2D or 3D color graphics with windows based software. It can be used on rough or curved surfaces for testing. Storage capacity of up to 20,000 records in the handheld computer, so its reliable evaluation of reinforcement corrosion in anaerobic concrete environment.



CANIN + Half Cell

The Canin+ is the fastest instrument for corrosion analysis, offering a practical, cost-effective approach to the investigation of reinforced concrete. Corrosion analysis with the Canin+ allows a rapid, comprehensive test of the site and provides a fast assessment of locations where corrosion is likely to take place. Unit includes: Canin+ indicating device , adjustable, padded carrying strap, protection sleeve for display instrument, transfer cable, USB-serial adapter, operating instructions, carrying case, Canin+ rod electrode with spare parts, electrode cable, bottle with copper sulphate (CuSO₄) 250g, Canin ProVista PC software on memory stick.

Resipod Resistivity Meter

Resipod is a fully integrated 4-point Wenner probe, designed to measure the electrical resistivity of concrete in a completely non-destructive test. Operating on the principle of the Wenner probe, the Resipod is designed to measure the electrical resistivity of concrete or rock. The calculated resistivity depends on the spacing of the probes. It is extremely fast and stable and packaged in a robust, waterproof housing designed to operate in a demanding site environment. This is one of the most versatile NDT methods for concrete.



Giatec Xcell™

Giatec XCell™ is a novel tablet/smartphone-based NDT probe for fast, accurate and efficient detection of in-situ analysis of corrosion in reinforced concrete structures. It benefits from an advanced Bluetooth-enabled, maintenance-free sensor that measures the corrosion potential and sends it wirelessly to a tablet for generating half-cell contour plots in real-time. The results are analyzed using the Android-based application on site for the identification of locations with high probability of corrosion. The measured potential values are indicative of corrosion probability. Three models of the Giatec XCell™ are available.

Giatec Surf™ Surface Resistivity

It is a laboratory non-destructive test device for rapid, easy and accurate measurement of the surface electrical resistivity of concrete based on the four-probe (Wenner-Array) technique. Surf™ has a unique and customized setup, which enables the measurement of electrical resistivity with high accuracy by reducing the duration of the test and minimizing ambient effects. Surf™'s patented technology automatically takes measurements at each of the sensor points around the cylinder and then repeats the process for all 8 measurements in less than 15 seconds.



CorroWatch :- CorroWatch is a monitoring system for early warning of the onset reinforcement corrosion. It can also be used to estimate the time before corrosion of reinforcement begins so that corrective measures may be taken early enough to minimize repair costs.

The CorroWatch is a multi-probe device consisting of four black steel bars acting as anodes and a noble metal as the cathode. The bars are positioned at different elevations and when cast into concrete, CorroWatch allows determination of corrosion activity as a function of cover distance.

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ERE-Probe (Embeddable Reference Electrode)

The ERE-Probe (Embeddable Reference Electrode) is a stable, long life reference electrode for monitoring the half-cell potential of reinforcement. It is based on a manganese dioxide electrode in a

steel housing with a chloride-free, alkaline gel and having a porous cement plug at one end.

The ERE-Probe is a reference half-cell that is embedded in concrete and can be used to monitor the reinforcement potentials in critical areas for corrosion such as construction joints, splash zones in marine structures, bridge decks, and bottoms of bridge columns exposed to ingress of chlorides.



Deep Purple and Rainbow Indicator

Deep Purple and Rainbow Indicator are used to determine the depth of carbonation in samples of field concrete. Carbonation depth can be used to evaluate the cause of corrosion when conducting corrosion survey, to estimate service life where penetration of the carbonation front is critical and to make a rough estimate of concrete strength from the age of concrete and the relative humidity. The natural alkalinity of cement paste in concrete results in a protective oxide coating on steel reinforcement that prevents the steel from rusting.



Chloride and Alkali Content

RCT and RCTW

The RCT and RCTW systems are used to accurately and quickly determine the chloride ion content from powder samples of concrete obtained on-site or in the laboratory using the Profile Grinder or other means. The test results can be used for: Establishing the chloride ion profile for service life estimation & Diagnosing a structure for corrosion activity, in combination with other test systems such the Mini Great Dane, the GalvaPulse, and the Rainbow Indicator. The RCT (Rapid Chloride Test) is used to determine the amount of acid-soluble chlorides & The RCTW (Rapid Chloride Test Water) is used to determine the amount of water-soluble chlorides.



RAT (Rapid Alkali Test)

The RAT (Rapid Alkali Test) measures the amounts of sodium and potassium ions that may contribute to alkali-silica reaction (ASR) if reactive aggregates are present. ASR leads to expansive products that can cause extensive cracking in concrete structures.

The RAT measures the amount of sodium and potassium ions in the fresh concrete or in its constituents. The test may also be used for testing powder samples of hardened concrete.



Durability Assessment

PROOVE'it – Rapid Chloride Ion Permeability Test

The PROOVE'it system is used to evaluate the resistance of concrete to the ingress of chloride ions in three ways: By determining the total electrical charge that passes through a saturated concrete specimen by applying an electrical potential across the specimen in accordance with AASHTO T 277 or ASTM C1202. This is known as the "Coulomb Test" or the "Rapid Chloride Permeability Test (RCPT)." By measuring the penetration depth of chloride ions, after an electric potential has been applied to the specimen in accordance with Nordtest Build 492 to determine the "Chloride Migration Coefficient," which can be used to estimate the chloride diffusion coefficient for service life calculations. By measuring the current passing through a saturated concrete specimen and determining the bulk conductivity in accordance with ASTM C1760.



PROOVE'it is the computer-controlled microprocessor power supply and the Windows based software for testing and report preparation offers the testing up to 8 cells simultaneously & Cyclic testing option for effect of curing duration.

Concrete & Cement



Profile Grinder

The Profile Grinder is used to obtain concrete powder samples by precision grinding at small depth increments for accurate determination of the chloride ion profile for the following immersion of specimens in a chloride solution in the laboratory, e.g., according to NT Build 443 "Concrete, Hardened: Accelerated Chloride Penetration" or ASTM C1556 "Test Method for Determining the Apparent Chloride Diffusion Coefficient of Cementitious Mixtures by Bulk Diffusion," or On-site on structures that have been subjected to chloride ion ingress.

A grinding bit, 18 mm in diameter, grinds the concrete to a fine powder at exact depth increments, which can be selected between 0.5 mm to 2.0 mm. The bit is attached to a grinding machine that is held against the surface by a grinding plate. Grinding is accomplished by rotating the grinder within the grinding plate so that the bit removes a circular portion of the surface.

Merlin Bulk Conductivity

Merlin is used to measure the bulk electrical conductivity, or its inverse, the bulk electrical resistivity, of saturated 100 mm diameter concrete cylinders or cores with lengths up to 200 mm. The test is simple to perform and a measurement is obtained within two seconds. The conductivity of a saturated concrete specimen provides information on the resistance of the concrete to penetration of ionic species by diffusion.

Merlin can be used for the research and development to characterize the influence of new materials on the electrical conductivity of concrete. Optimizing mixture proportions and blends of supplementary cementitious materials to increase concrete service life. Evaluation of in-place concrete (using cores).



GWT (Germann Water permeation Test)

The GWT (Germann Water permeation Test) is used for on-site evaluation of the water permeation of the skin-concrete in finished structure and masonry panels. The GWT measures the permeation of water into the test surface under an applied pressure.

With the use of the optional laboratory kit, the GWT can be used to determine the water penetration characteristics of alternative concrete mixtures or surface sealers. GWT can be used for in-place testing to demonstrate that the concrete in the structure conforms to the water penetration requirements. re located at the front of the apparatus while the three graduated burettes for measuring water volume are mounted on the upper panel

Field Water Impermeability Apparatus

Used to determine the impermeability of concrete to water. The test is performed by placing the sample in the special chamber (measuring 250 x 250 x 220 (h) mm, height being adjustable) and securing it between the upper flange and the lower gasket delimiting the test surface.

Water under pressure is then applied to the surface (dia. 100mm) for the duration prescribed by the Standard. A precision valve controls water pressure shown by the dial gauge, pressure being supplied by an air compressor (10 bar). The apparatus can be used for testing three samples at a time. All parts coming into contact with water are in stainless steel. Inlet-outlet taps are located at the front of the apparatus while the three graduated burettes for measuring water volume are mounted on the upper panel



Rapid, Freeze-Thaw Cabinet

The Rapid Freeze-Thaw Cabinet is used to measure the resistance of concrete to deterioration caused by repeated cycles of freezing and thawing in water. The system is designed to test up to eighteen 3" x 4" x 16" (76 x 102 x 406cm) concrete specimens simultaneously, with one being a control. It allows users to establish field control using correlations between concrete strength and durability. It is useful in the evaluation of the durability of aggregates, as well as the properties of admixtures.

Concrete & Cement

Sonometer

The Sonometer determines changes in resonant frequency of concrete specimens subjected to alternate cycles of freezing and thawing with the Humboldt Freeze-Thaw Cabinet. This apparatus closely follows design parameters set up over 40 years ago by the Portland Cement Association research laboratories. The original PCA design has been modified by changing to solid state circuitry and addition of a built-in cathode ray oscilloscope. The oscilloscope confirms that peak reading on meter is actual resonance and not a harmonic.



E-Meter for Flexural Resonance of Concrete

The E-Meter can determine flexural resonance of concrete under accelerated freezing and thawing cycles and aggressive environments. It determines the resonant frequencies of the three modes of vibration and is the only method of calculating the following material parameters non destructively: such as Youngs Modulus of Elasticity, Modulus of Rigidity, Poissons Ratio and Damping Constant. Frequencies are automatically scanned in one of four ranges. A semi-automatic feature facilitates the fast identification of resonance.

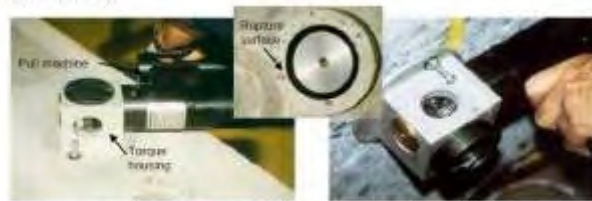
Repair Quality

Bond Test

The BOND-TEST is used to conduct a pull-off test in accordance with ASTM C1583, "Test Method for Tensile Strength of Concrete Surfaces and the Bond Strength or Tensile Strength of Concrete Repair and Overlay Materials by Direct Tension (Pull-off Method)." The obtained pull-off strength can be used to evaluate the in-place bond strength between a repair overlay and the substrate. It is useful to evaluate the in-place tensile strength of concrete or other materials. As well as to evaluate the effect of surface preparation procedures on the tensile strength of the substrate before applying a repair material or overlay.



Testing Example



Testing with the TORQ-TEST on a bridge girder before application of CFRP strips

Torq Test

he TORQ-TEST is used to evaluate the shear strength of Surface concrete, the bond between carbon fiber reinforced polymer (CFRP) sheets or strips and concrete, the bond between an overlay and concrete.

A disc with an integral ring, 55-mm inner diameter and 75-mm outer diameter, is bonded to the surface using a rapid curing adhesive (GRA) and is used to impart shearing stress to the test surface.

DSS Test

The DSS-TEST is used to measure the direct shear bond strength of a carbon fiber reinforced polymer (CFRP) laminate bonded to concrete. The effectiveness of the CFRP laminates to act as external reinforcement relies on adequate shear bond strength with the concrete.

CFRP laminates are used as external reinforcement to strengthen existing concrete structural elements. The laminates are bonded to the concrete with epoxy adhesives. The strip is bonded perpendicular to the edge of the concrete so that it extends 200 mm beyond the edge.



Power Test

The POWER test system is used for proof-load testing of anchors in concrete. An adaptor of the appropriate size is threaded to the anchor bolt. A counter pressure assembly is placed on the concrete so that it is centered with the anchor bolt. A pull bolt with coupler is threaded into the adaptor and attached to a hydraulic pull machine, which is the same as used for pull-out testing. The load is increased to the required proof load and held for at least 10 seconds. If the load is maintained during the hold period, the anchor passes the proof load test.

Concrete & Cement

Cement

ASTM-Compliant Mixer

ASTM-compliant mixer for mixing hydraulic cement pastes and mortars of plastic consistency. Mixer includes H-3844 Bowl Positioning Adapter, a 5-qt. (4.73L) stainless steel bowl and 1 flat, stainless steel beater for mixing heavy materials. Hobart Model No. N-50 operates on principle of planetary action—beater reaches every part of the batch, rotating on its axis in opposite directions as it moves around the bowl. Thoroughly blends, mixes and aerates all ingredients for consistent, predictable finished batches.



Mortar Mixer w/ Program Control and Sand Feed

Mixer with 5L (5.3qt) capacity provides automatic program control and sand feed with manually-controlled cement and water supply. Two-speed mixer designed to mix cement mortars and pastes. Mixer features a 5 Liter (5.28qt) stainless steel bowl and ASTM stirrer, and a two speed motor, which runs at either 140 ± 5 rpm or 285 ± 10 rpm. Heavy-duty constructions of aluminium and cast iron provides durability and a long service life. The Mixer design conforms to CE standards, featuring a mixing bowl cover, electronic position monitoring of the mixing bowl, emergency stop switch and restart protection.

Length Comparators

Length comparators measure length changes of hardened cement paste, mortar and concrete prismatic specimens. Indicators are mounted on a sturdy upright support attached to a solid triangular base. All units include stationary and movable anvils designed to fit H-3260 gauge studs, which are cast into test specimens and an invar reference bar.



Cement Autoclave

Uses accelerated means of estimating delayed expansion of Portland cement caused by hydration of CaO and MgO. Test bars are exposed to controlled steam pressure and corresponding constant temperature. Unit produces 60-350 psi (0.4-2.4MPa) range of pressures and consists of steam vessel, pressure regulator, pressure gauge, air vent valve, power switches and safety pop valve set at 350psi. Includes thermometer, wrench and 5 gaskets. Additional gaskets, heating units and safety pop valves are available as replacement parts.

Vicat Apparatus, Computer Controlled

The Computer-controlled, Vicat Apparatus can be used to determine the initial and final setting time of cement, mortar pastes and gypsum. This six-station apparatus automatically monitors the setting process and provides output curves of the process. The test sequence is programmable and tests can be performed in air or in the supplied, temperature-controlled water bath. The computer-controlled vicat apparatus includes: Base unit as a table model apparatus with a control unit and standard software for performing programmable test sequences and for data recording.



Gillmore Apparatus

Used to determine initial and final set times of Portland cement, masonry cement, hydraulic hydrated lime and certain mortars. Comprised of two stainless steel needles with 4.8mm cylindrical flat-end needles. One is 2.12mm dia. 1/4lb. (113.4g) weight for initial set. The second is 1/24" (1.06mm) dia., 1lb (453.6g) weight for final set.

Concrete & Cement

Cement Bleeding Apparatus

Used to determine bleeding rate and bleeding capacity of cement paste and mortar by direct and continuous procedure. Includes noncorrosive metal container for paste or mortar, collecting ring, support stand and necessary glassware. Rubber-covered double V-jaw burette clamp holds stopcock in position; single rubber-covered jaw clamp supports the burette and funnel assembly.



Cement Bleeding Apparatus



Water Retention Apparatus

The apparatus is used in specification tests of masonry cement and physical testing of quicklime and hydrated lime. Unlike older models, the newly designed unit incorporates a vacuum regulator and gauge system in place of the old mercury manometer and relief column. The complete unit consists of an aspirator pump, vacuum regulator, vacuum gauge, three-way stopcock, flask, rubber gasket, brass funnel, perforated brass dish, filter paper and hardwood stand.

Blaine Air Permeability Apparatus

Determines fineness of Portland cement in terms of specific surface expressed as total surface area in square centimeters per gram of cement. Consists of: calibrated U-tube manometer, ground glass joint, stainless steel test cell and plunger, rubber aspirator bulb and perforated disc. Includes 8 oz (226.8g) bottle of red manometer fluid, filter paper, wood block for holding test cell during filling and funnel. Mounted on finished wood panel with rubber-footed base. Electronic and computer controlled Blaine Air permeability (Dyckerhoff) apparatus are also available.



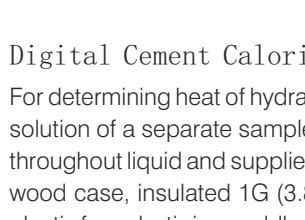
Organic Impurities Test Set

Determines presence of injurious organic compounds in sands used in cement mortar or concrete. Test serves as warning that further tests of sands are necessary before they can be approved for use. Air shipments must meet Dangerous Goods requirements because of Sodium Hydroxide Beads. Order H-3493X without Sodium Hydroxide Beads to avoid Dangerous Goods requirements.



Wagner Turbidimeter

Determines fineness of Portland cement, using photoelectric cell to measure light passing through suspended pulverized material. Microamp meter measures current generated in the cell; indicated reading is measure of turbidity of the suspension. Includes photoelectric cell (and light source in metal cabinet, timing burette and stand, wet sieving assembly including gauge and spray nozzle, microamp meter, 3 flasks, 4 test tubes, stirring apparatus and instruction book.



Digital Cement Calorimeter

For determining heat of hydration of cements by measuring difference between heat of solution of dry cement and heat of solution of a separate sample partially hydrated for 7 to 28 days. Constant-speed stirrer maintains uniform temperature throughout liquid and supplies sufficient agitation to keep solid reactant suspended in the acid mixture. Includes insulated wood case, insulated 1G (3.8L) can; 1 pt. (0.47L) vacuum jar with stopper; 2-channel, Precision Digital Thermometer; plastic funnel; stirring paddle and chuck; geared synchronous motor.



Marsh Funnel Viscometer

The Marsh Funnel Viscometer is a rugged, easy to operate instrument that is used for making rapid, on the spot measurements of drilling mud viscosity. Marsh Funnel readings are only general measurements, but the frequent reporting of the Marsh Funnel Viscosity will alert the mud engineer to sudden changes in the mud viscosity that could require corrective action. The Marsh Funnel Viscosity is the ratio of the speed of the mud as it passes through the outlet tube (the Shear Rate) to the amount of force—the weight of the mud itself, which is causing the mud to flow (the Shear Stress).

Mud Balance

The Mud Balance provides a simple, practical method for the accurate determination of fluid density. The item's durable construction makes it ideal for field use. It's high-impact plastic case protects the balance during transport while providing a secure base for the balance during use. The H-4790 Mud Balance meets all the requirements of the API standard procedures for testing water base drilling fluids, oil base drilling fluids and oil well cements.

